

Confidence Calibration for Multimodal Indoor Lower-Limb Activity Recognition under Sensing Degradation

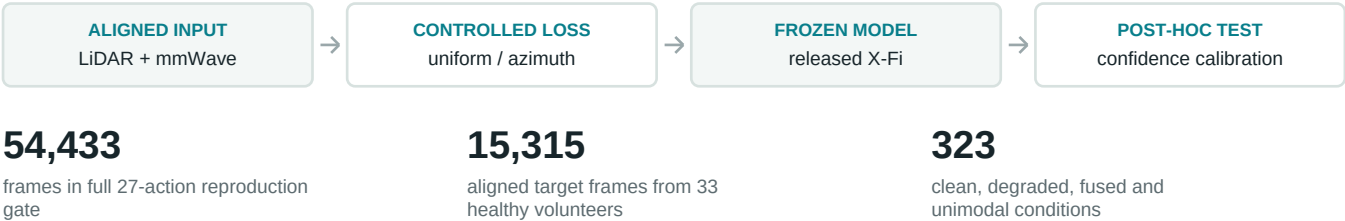
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RESEARCH QUESTION

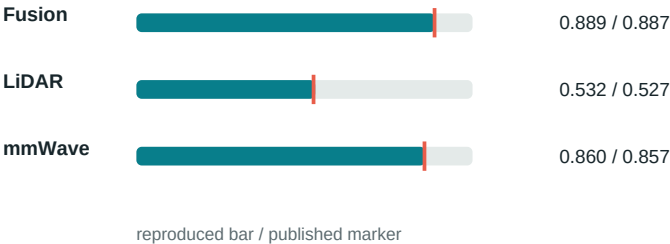
When aligned LiDAR and mmWave point observations deteriorate, can observable quality cues make a frozen X-Fi model's confidence more trustworthy than one pooled temperature?

STUDY DESIGN



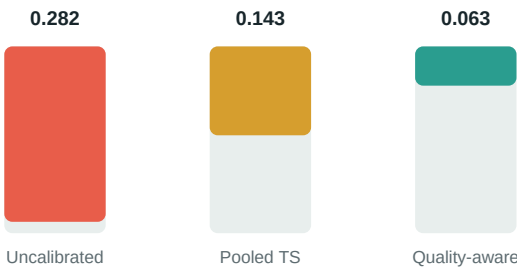
1. REPRODUCTION GATE

The released checkpoint was first checked on the complete clean validation cohort. All three masks were within the frozen 0.03 tolerance.



2. CONFIDENCE RELIABILITY

Mean Expected Calibration Error (ECE) across 240 degraded fused test conditions. Lower is better; temperature scaling does not alter class predictions.



WHAT THE EXPERIMENT SHOWS

- Recognition degraded sharply: fused Accuracy was 0.900 on clean target data and averaged 0.482 across degraded fused conditions. Under severe asymmetric degradation, fixed fusion could underperform the stronger available unimodal branch.
- Quality-aware temperature scaling reduced mean ECE from 0.143 to 0.063 and NLL from 1.968 to 1.846 versus pooled scaling. Calibration improved probability reliability, not recognition Accuracy; both methods retained the same predictions.

RESEARCH IMPLICATION

Observable sensing quality can improve how much a frozen multimodal model should be trusted, but cannot repair a wrong prediction. A focused dissertation extension is quality-gated fusion or selective abstention, followed by tests under real sensor faults.

CLAIM BOUNDARIES

Healthy volunteers; controlled software point loss; one released frozen checkpoint; no rehabilitation-patient, clinical, physical-failure, or real-time deployment validation.

Sources: MM-Fi (NeurIPS 2023); X-Fi (ICLR 2025)

<https://github.com/zyyo1999-ship-it/xfi-multimodal-reliability>